

Vishal Vetcha

[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

+91 8451973398 | vishal8451973398@gmail.com | Visakhapatnam, Andhra Pradesh, India

Education

B.Tech, Computer Science & Engineering

(Aug 2023 – Present)

Gandhi Institute Of Technology And Management (GITAM), Visakhapatnam

CGPA: 9.21 | SGPA: 10 (SEM 6)

Munich International Summer University, AI and ML in Corporate Finance

(Jul 2026 – Aug 2026)

LMU Munich, Germany | Grade: 1.3 / ECTS (European) Grade A (Highest Distinction)

Skills

Programming Languages: Python, C, Java

AI/ML: PyTorch, TensorFlow, Deep Neural Networks, OpenCV, LangGraph, RAG

Data Engineering: Data Pipelines, ETL, Apache Kafka, REST APIs

Databases: MySQL, PostgreSQL

Cloud, DevOps & Tools: AWS Cloud Fundamentals, Microsoft Azure, Azure DevOps, Docker, Git

Core: Data Structures & Algorithms, Software Development, Operating Systems, Computer Networks

Professional Experience

Reliance Industries Limited – Artificial Intelligence Intern ([Certificate](#))

(May 2026 – Jun 2026)

Proposed and architected a custom Core Machine Learning model solution capable of **reducing enterprise token usage by approximately 25%**, optimizing inference efficiency and operational cost. Engineered **production-oriented AI and edge-inference pipelines** integrating REST APIs and optimized computer vision workflows for **CPU-only, resource-constrained environments with as low as 4GB RAM**.

Amazon ML Summer School 2026 ([Certificate](#))

(Jul 2026 – Aug 2026)

Selected among the **top 2% from 130,000+ applicants nationwide**. **Core Topics:** Supervised Learning, Deep Neural Networks, Unsupervised Learning, Sequential Models, Causal Inference, Generative AI.

PwC (PricewaterhouseCoopers) – Launchpad Trainee ([Certificate](#))

(Feb 2026 – Aug 2026)

Selected for an intensive training program focused on advanced technical frameworks and strategic consulting methodologies for solving industry-level business challenges.

GitHub Community – Core Member

(Aug 2025 – Present)

Collaborating on deep neural network architectures, conducting technical workshops, and contributing to the development of AI/ML projects.

Projects

MicroHawk – Edge-Deployable Agentic AI Drone System

[Project Repository](#) | [Demo](#)

Architected an **edge-compatible autonomous drone platform** integrating a local **3.8B Qwen2.5-VL agent**, voice control, RGB vision, telemetry, deterministic safety, and Unity physics via Python-C#, **8 FastAPI endpoints**, and persistent WebSocket communication. Engineered a **50 Hz flight controller** with **10 Hz telemetry**, altitude/geofence/speed/battery failsafes, return-to-home, typed Pydantic plans, and **86 automated Python/Unity tests** across the autonomy stack.

Demonstrated structural inspection and restricted-zone patrol with **640×480 vision evidence**, temporal dwell detection, autonomous investigation, evidence capture, replanning, and physical landing; enforced an **LLM safety boundary**.

Brain Imaging Analysis – Tumor & Hemorrhage Classification

[Springer Nature Publication](#)

Architected a novel **CNN-based deep learning pipeline with a custom multi-view attention head** for brain tumor and hemorrhage detection, classification, and explainable localization.

Trained and optimized the system on **10,000+ medical images**, achieving **99% classification accuracy**; integrated Grad-CAM explainability and conducted extensive ablation studies.

FreshConnect – Smart India Hackathon Grand Finalist

[Project Repository](#)

Developed a **Flutter-Firebase** agri-tech platform connecting farmers directly with consumers through real-time listings, orders, and transactions. Engineered a **Bio-Waste Management System** for agricultural waste monetization; led the team to the **Smart India Hackathon National Grand Finale (Top 5)**.

Research Publications

Pathology-Adaptive Multi-View Attention for Unified Brain Imaging Analysis

International Conference on Frontiers in Advanced Computing and Emerging Intelligent Technologies (FACEIT), Springer Proceedings, Scopus Indexed.

Paper presented at **National Institute of Technology Warangal, India, 2026**. [Springer Nature Link](#)

Certifications & Achievements

PCEP – Python Certified -83%

CLE - C Certified -90%

Top 20 International Cohort – LMU Munich International Summer University, Germany - Grade A (Highest)

Amazon ML Summer School 2026 - Top 2%

Smart India Hackathon 2024 – National Level Grand Finalist (Top 5)

Bronze Medal – Stanford University RNA 3D Folding Competition

PwC Launchpad Trainee

Entrepreneurship Award – Venture Development Centre (GITAM)

IdeaBlast 2024 – Finalist (Top 10)